ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

The pixel pitch is 0.005 mm, so f_x = 16/0.005 = 3200 pixels. Sensor width = 6.4 mm. Belt field of view = 400x6.4/16 = 160 mm; spatial resolusion = 160/1280 = 0.125 mm/pixel.

Q1(b) [3 marks]

ADC step = 5/2^12 = 1.2207 mV. Since sensitivity is 10 mV/N, ΔF = 1.2207/10 = 0.122 N. I report this as 0.122 mN.

Q1(c) [3 marks]

For zero-order hold, F = e^(AT) and G = integral_0^T e^(Aτ)B dτ. Here:

$$A = -2, B = 1, T = 0.1s$$

Thus:

$$F = e^(- 0.2) = 0.818731$$

$$G = (1 - e^(- 0.2))/2 = 0.0906346$$

Therefore:

$$x_{k+1} = 0.818731x_k + 0.0906346u_k$$

With $AT = (-2)(0.1) = -0.2$, $F = e^{-0.2} = 0.8187308$. For G, $(1 - F)/2 = (1 - 0.8187308)/2 = 0.1812692/2 = 0.0906346$. Thus the input contribution is positive.

Question 2 [10 marks]

Q2(a) [2 marks]

Choose $x_1 = \theta$ and $x_2 = \dot{\theta}$. Then $\dot{x}_1 = x_2$ and $\dot{x}_2 = [\tau - b\, x_2 - mg\ell \sin(x_1)]/J$. At equilibrium:

$$x_2 = 0 and_0 = mg\ell sin\theta_e$$

Since |sin θ_e| less than or equal to 1, an equilibrium exists when $|\tau_0|$ less than or equal to $mg\ell = 2.4525$ N m.

At rest $\dot{\theta} = 0$ and $\ddot{\theta} = 0$, so the damping term vanishes. Rearranging gives $\tau_0 = mg\ell \sin\theta_e$. Here $mg\ell = 1 \times 9.81 \times 0.25 = 2.4525$ N m. Since $-1 \le \sin\theta_e \le 1$, the required torque must lie between -2.4525 and $+2.4525$ N m.

Q2(b) [2 marks]

With x = [θ, $\dot{\theta}$]^T, A = [[0,1], [-(mgℓ/J)cos θ_e, -b/J]], B = [[0],[1/J]]. At θ_e = 60°, A = [[0,1], [-61.3125, -2.5]] and B = [[0],[50]].

At the equilibrium, the constant torque balances gravity. For a small perturbation, use $\sin(\theta_e + \delta\theta) \approx \sin\theta_e + \cos\theta_e\,\delta\theta$. The constant terms cancel against τ_0; the remaining linear term gives the local state matrix.

Q2(c) [4 marks]

At rest, $\dot{\theta} = \ddot{\theta} = 0$, so the torque balance is Kp(θd-θss) = mgℓ sin θss. With Kp = 50, θd = π/2 and mgℓ = 2.4525, solve 50(π/2-θss) = 2.4525 sin θss. This gives θss approximately 1.52181 rad = 87.193°, so the error is 90°-87.193° = 2.807°; it is not zero.

Rearranging the steady-state equation gives $50(\pi/2 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$. I used radians in the equation. Solving numerically gives $\theta_{ss} \approx 1.522$ rad $\approx 87.19°$, so the gap from $90°$ is about $2.81°$.

Q2(d) [2 marks]

Question 3 [10 marks]

Q3(a) [2 marks]

Finite differencing amplifies encoder noise, making the velocity estimate noisy.

Q3(b) [2 marks]

$$y[k] = 0.8818y[k-1] + 0.0591x_k + 0.0591x[k-1], DCgain1$$

Q3(c) [4 marks]

With L=[ℓ_1,ℓ_2]^T and C=[1,0], A-LC=[[-ℓ_1,1], [-ℓ_2,-2]]. Its characteristic polynomial is det(sI-(A-LC)) = (s+ℓ_1)(s+2)+ℓ_2 = s^2+(ℓ_1+2)s+(2ℓ_1+ℓ_2). Desired poles at -10 give $(s+10)^2$=s^2+20s+100. Hence ℓ_1+2=20 gives ℓ_1=18, and 2ℓ_1+ℓ_2=100 gives ℓ_2=64.

Expanding $(s+\ell_1)(s+2)+\ell_2$ gives $s^2 + (\ell_1+2)s + (2\ell_1+\ell_2)$. The desired polynomial is $(s+10)^2 = s^2 + 20s + 100$. Matching the s coefficient gives $\ell_1 = 18$; matching the constant term then gives $36 + \ell_2 = 100$, hence $\ell_2 = 64$.

Q3(d) [2 marks]

An observer can estimate velocity from position measurements. A filter can be safer because it smooths noise.

Question 4 [10 marks]

Q4(a) [2 marks]

Closed-loop characteristic polynomial is $s^2+(a+bk_2)s+(c+bk_1)=s^2+4s+8$, giving poles $-2\pm2j$. At DC, $\theta/r = bN_{ff}/(c+bk_1)$, so unity gain requires $N_{ff}=(c+bk_1)/b=(6+2)/1=8$.

Comparing coefficients with $s^2 + 4s + 8$ gives $a + bk_2 = 4$ and $c + bk_1 = 8$. The feedforward gain is chosen from the nominal plant so the reference scaling is unity; here this gives $N_{ff} = 8$. The closed-loop roots of $s^2 + 4s + 8$ are $-2 \pm 2j$.

Q4(b) [2 marks]

$$\theta_{ss} = 0.75 \times 8/(6 + 2) = 0.75,$$

the percentage tracking error is 25%

Q4(c) [4 marks]

The characteristic polynomial is $s^3+(a+bk_2)s^2+(c+bk_1)s+bK_I$. I use the target $s^3+6s^2+12s+16$, giving $k_2=4$, $k_1=6$, $K_I=16$.

Q4(d) [2 marks]